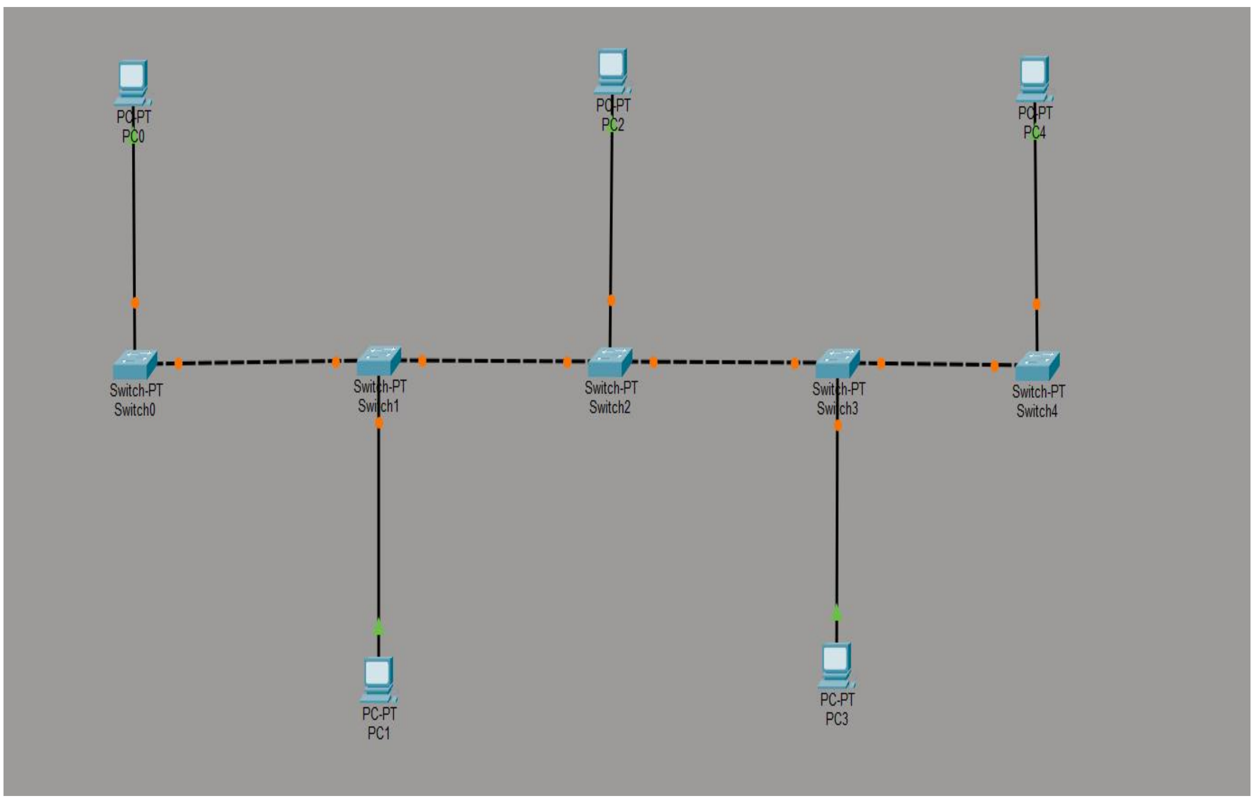
Lab Report-01

Problem: Bus Topology

Description: Bus topology is a network setup where all devices are connected to a single central cable (the bus). Data travels in both directions along the bus, but only one device can transmit at a time to avoid collisions. Terminators at both ends of the cable prevent signal reflection. It's inexpensive and easy to install for small networks but hard to troubleshoot and less efficient with high traffic or many devices.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1. First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image

2. Use automatic connecting cable to connect devices with others or individually use the copper straight through cable to connect the devices and use copper cable to connect the switches.

3. Configure the PCs with IPV4 Address and subnet mask according to the IP Addressing Table given below

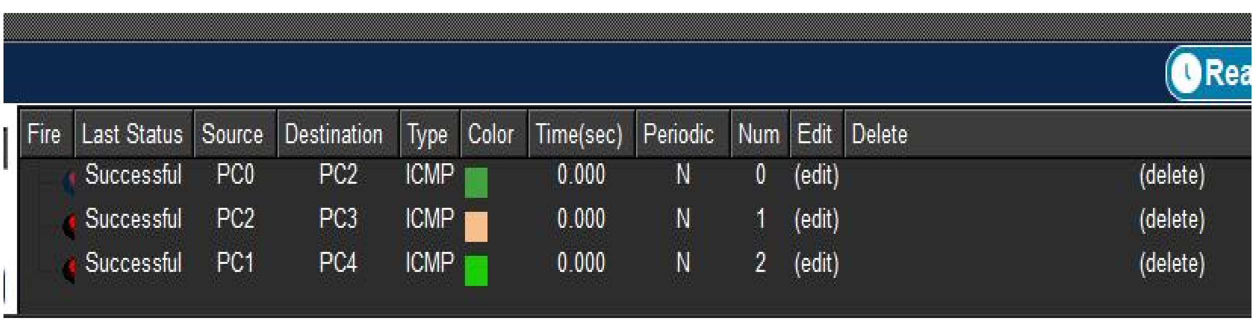
4. Set subnetmask(255.255.255.0)

5. Assign IP Address

    - PC0-192.168.10.1
    - PC1-192.168.10.2
    - PC2-192.168.10.3
    - PC3-192.168.10.4
    - PC4-192.168.10.5
    - PC5-192.168.10.6

6. Test Connectivity: Open a command prompt on one PC and use the "ping" command

7. Test File Sharing:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC2	PC3	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC1	PC4	ICMP		0.000	N	2	(edit)	(delete)

Conclusion: Bus topology is a straightforward and economical networking structure where all devices are connected to a single central cable. It is simple and cost-effective, ideal for small networks. It suffers from performance issues and is hard to troubleshoot as more devices are added. A single cable failure can bring down the entire network.

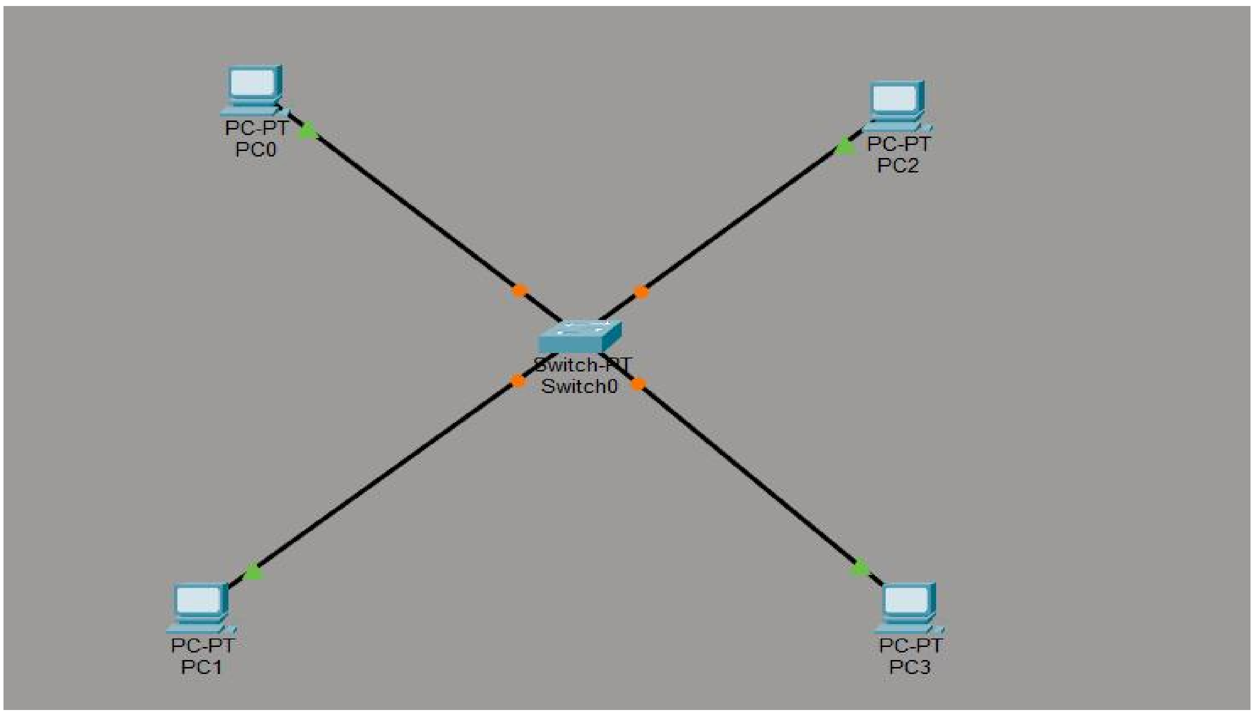
Lab Report-02

Problem: Star Topology

Description: Star topology is a network configuration where all devices are individually connected to a central device (such as a switch or hub). The central device manages and controls all data traffic. If one device fails, the rest remain unaffected, but if the central device fails, the entire network goes down. It offers better performance, easier troubleshooting, and is more scalable than bus topology but requires more cable and is costlier.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1.First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image

2.Use automatic connecting cable to connect devices with others or individually use the copper straight through cable to connect the devices and use copper cable to connect the switches.

3.Configure the PCs with IPV4 Address and subnet mask according to the IP Addressing Table given below

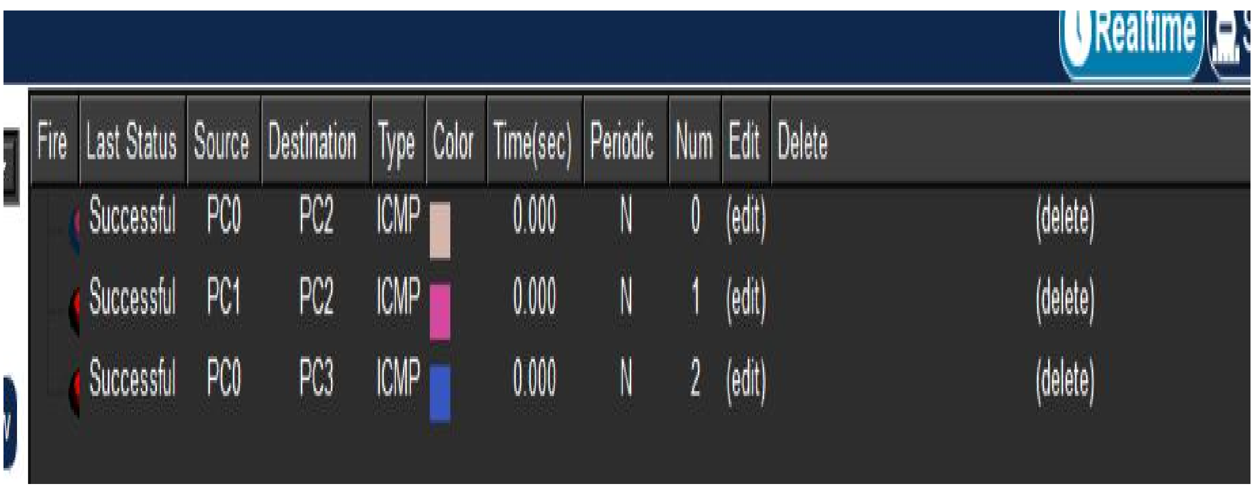
4.Set subnetmask(255.255.255.0)

5.Assign IP Address

- PC0-192.168.10.1
- PC1-192.168.10.2
- PC2-192.168.10.3
- PC3-192.168.10.4

6.Test Connectivity: Open a command prompt on one PC and use the "ping" command

7.Test File Sharing:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC0	PC3	ICMP		0.000	N	2	(edit)	(delete)

Conclusion: Star topology is a reliable and widely used network design due to its centralized structure. It offers easy troubleshooting, high performance, and minimal impact from individual device failures. The central hub or switch efficiently manages data flow between devices. However, the entire network depends on the central device—if it fails, the whole system is affected.

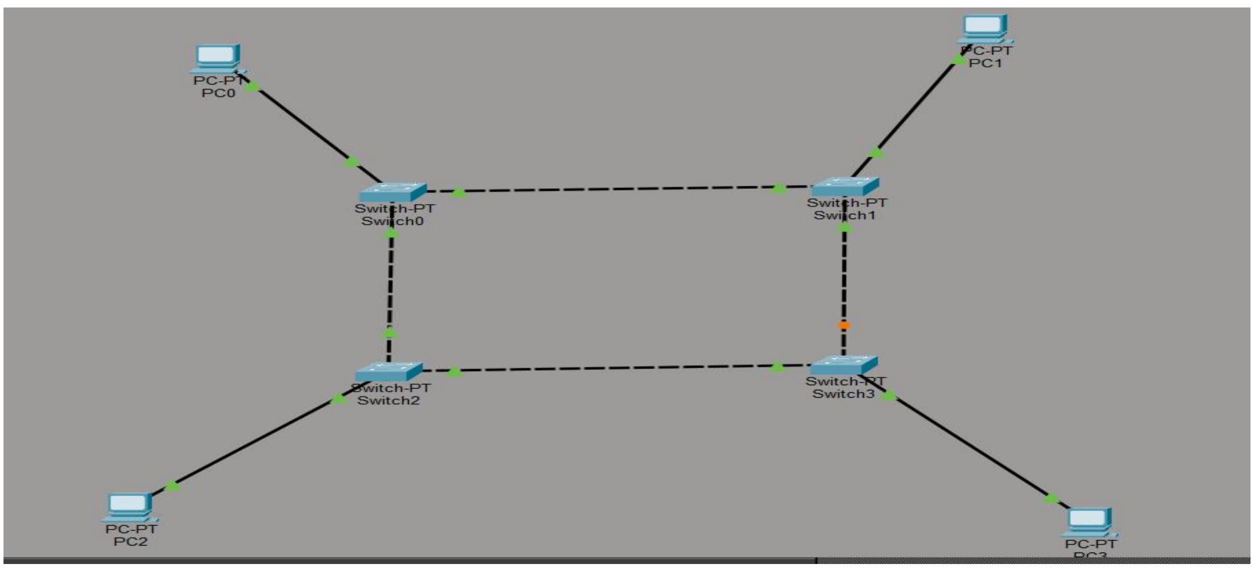
Lab Report-03

Problem: Ring Topology

Description: Ring topology is a network configuration where each device is connected to exactly two other devices, forming a closed loop or ring. Data travels in one direction, passing through each device until it reaches its destination. Each device acts as a repeater, boosting the signal to ensure it reaches the next node. It offers orderly data transfer and predictable performance but is vulnerable to failure—if one device or connection breaks, the whole network can be disrupted.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1.First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image

2.Use automatic connecting cable to connect devices with others or individually use the copper straight through cable to connect the devices and use copper cable to connect the switches.

3.Configure the PCs with IPV4 Address and subnet mask according to the IP Addressing Table given below

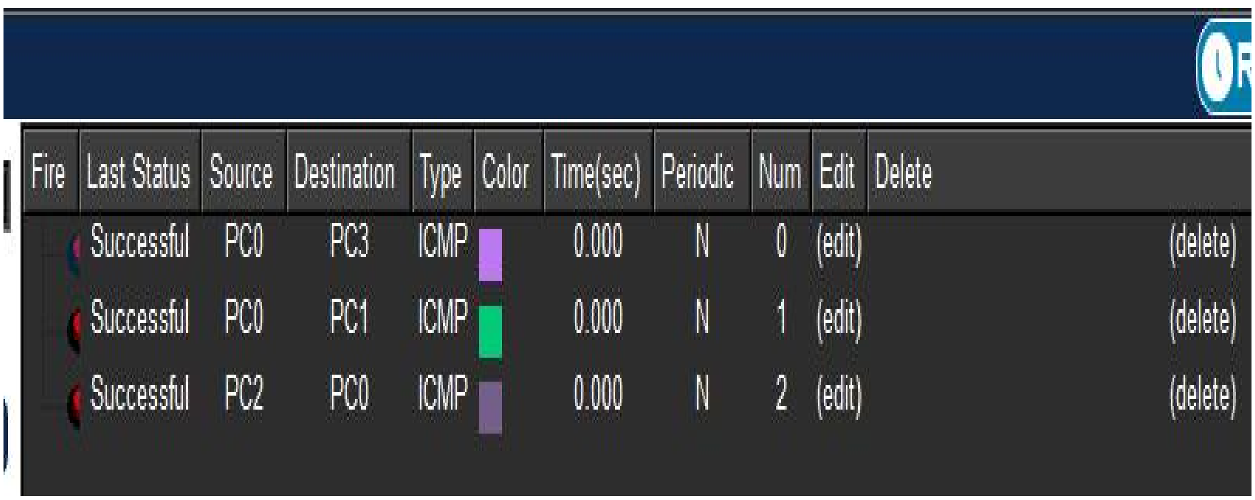
4.Set subnetmask(255.255.255.0)

5.Assign IP Address

- PC0-192.168.10.1
- PC1-192.168.10.2
- PC2-192.168.10.3
- PC3-192.168.10.4

6.Test Connectivity: Open a command prompt on one PC and use the "ping" command

7.Test File Sharing:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC0	ICMP		0.000	N	2	(edit)	(delete)

Conclusion: Ring Topology provides an organized and efficient way to transmit data with equal access for all devices.It ensures predictable performance under light to moderate loads.However the main drawback is its dependency on each connection if a single device or link fails,the entire network can be affected.Despite this using a dual ring can improve reliability and fault tolerance

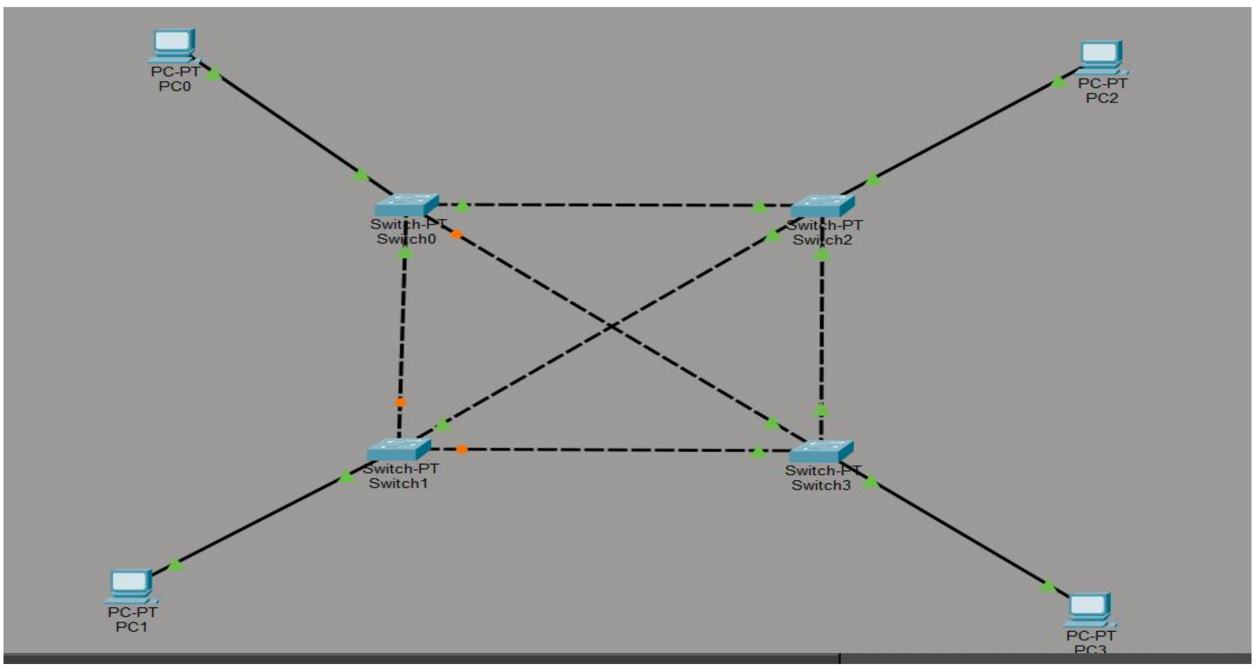
Lab Report-04

Problem: Mesh Topology

Description: Mesh topology each device to every other device in the network ,ensuring multiple paths for data transmission.This provides high reliability and fault tolerance as data can still flow even if one link fails.It is ideal for critical systems needing constant uptime.However it is costly and complex due to extensive cabling and configuration.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working procedure:

1.First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image

2.Use automatic connecting cable to connect devices with others or individually use the copper straight through cable to connect the devices and use copper cable to connect the switches.

3.Configure the PCs with IPV4 Address and subnet mask according to the IP Addressing Table given below

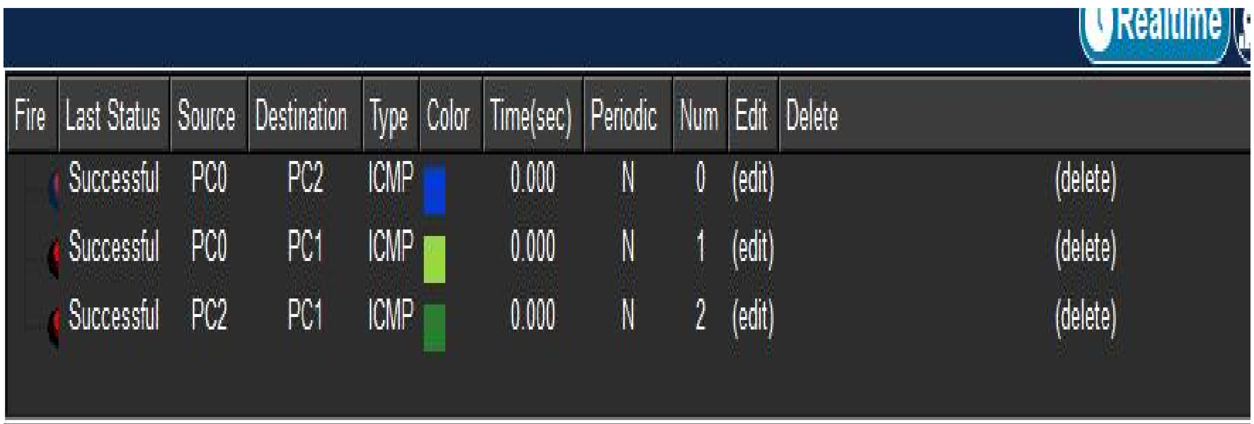
4.Set subnetmask(255.255.255.0)

5.Assign IP Address

- PC0-192.168.10.1
- PC1-192.168.10.2
- PC2-192.168.10.3
- PC3-192.168.10.4

6.Test Connectivity: Open a command prompt on one PC and use the "ping" command

7.Test File Sharing:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC1	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC1	ICMP		0.000	N	2	(edit)	(delete)

Conclusion:

Mesh topology offers high reliability and redundancy by providing multiple paths between devices, ensuring that even if one connection fails, data can take an alternate route. It is ideal for critical systems where continuous network availability is essential. However, it is complex to install, requires more cabling, and is costly to maintain. Despite the expense, its robustness makes it suitable for large, mission-critical networks.

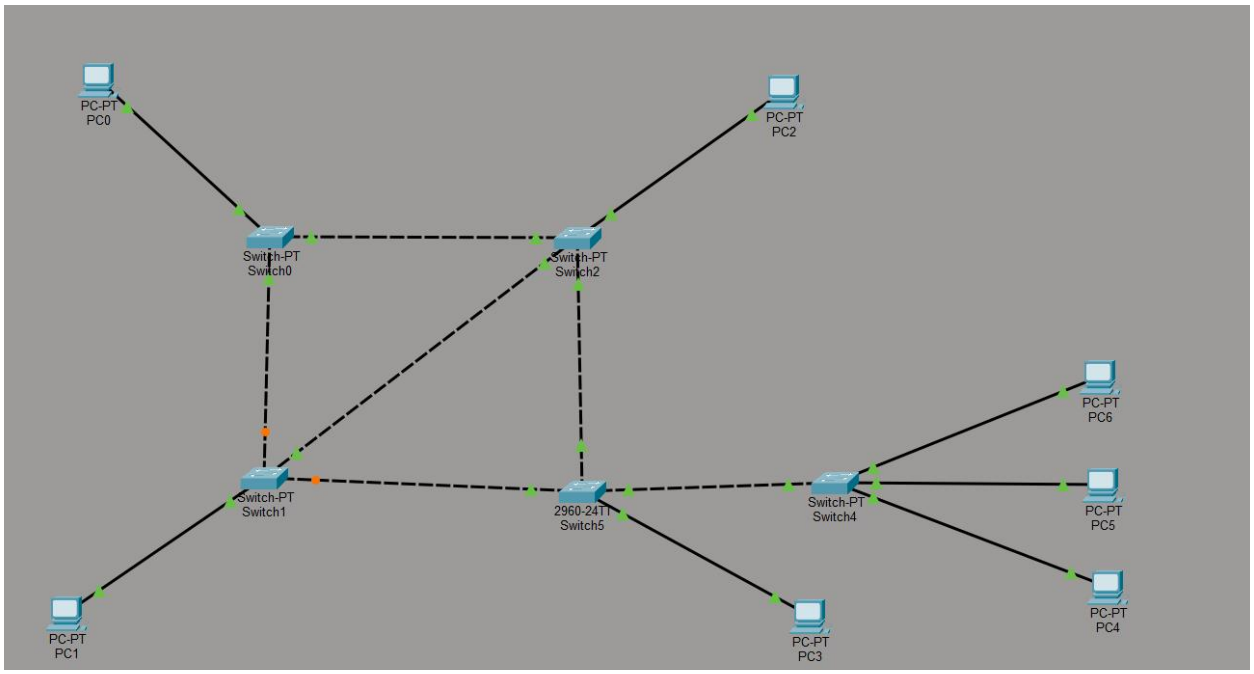
Lab Report-05

Problem: Hybrid Topology

Description: Hybrid topology combines two or more basic topologies (like star, bus, or ring) to create a flexible and scalable network. It benefits from the strengths of each topology used, offering better performance and reliability. This makes it ideal for large, complex systems. However, it can be expensive and difficult to manage.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1.First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image

2.Use automatic connecting cable to connect devices with others or individually use the copper straight through cable to connect the devices and use copper cable to connect the switches.

3.Configure the PCs with IPV4 Address and subnet mask according to the IP Addressing Table given below

4.Set subnetmask(255.255.255.0)

5.Assign IP Address

- PC0-192.168.10.1
- PC1-192.168.10.2
- PC2-192.168.10.3
- PC3-192.168.10.4
- PC4-192.168.10.5
- PC5-192.168.10.6

6.Test Connectivity: Open a command prompt on one PC and use the "ping" command

7.Test File Sharing:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0	PC3	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC1	ICMP		0.000	N	2	(edit)	(delete)

Conclusion: Hybrid topology is a highly flexible and scalable network design that combines the strengths of multiple basic topologies like star, bus, and ring. It offers enhanced performance, reliability, and adaptability, making it ideal for large and complex network environments. Although it can be costly and complex to design and manage, its ability to meet diverse networking needs makes it a preferred choice for modern organizations.

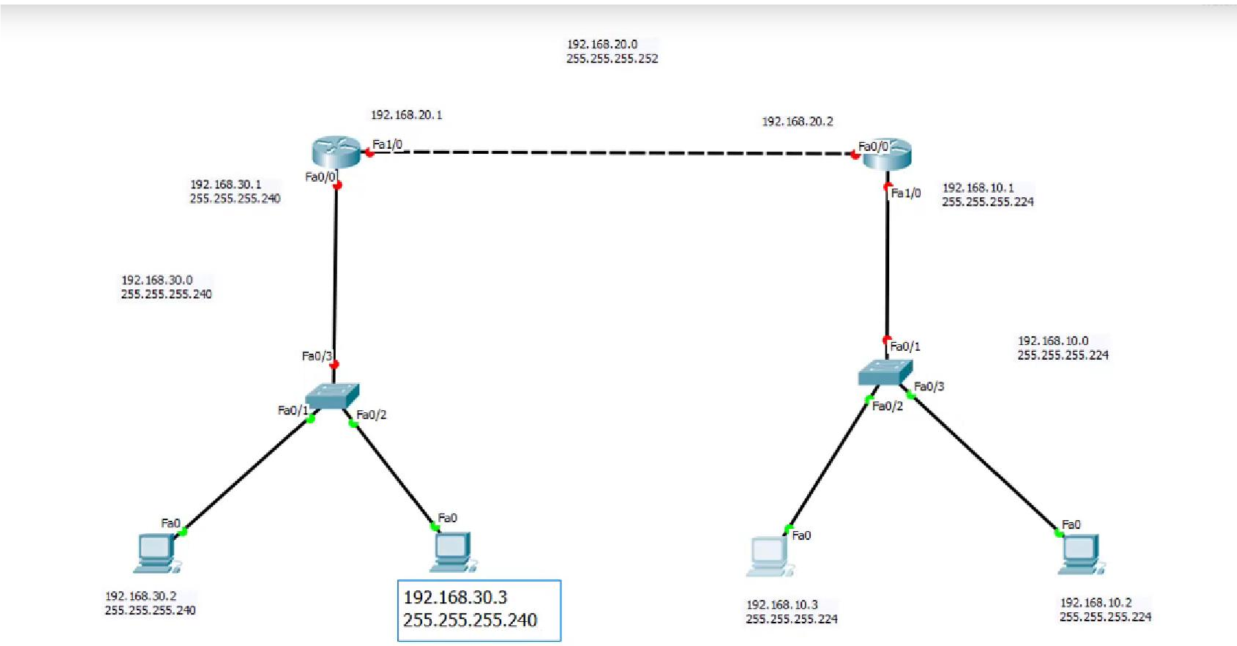
Lab Report-06

Problem: Static Routing

Description: Static routing is a method of routing where routes are manually configured and maintained by a network administrator. Each route must be explicitly defined, specifying the destination network and the next-hop address. It is simple and secure, making it suitable for small or stable networks. However, it lacks adaptability—changes in the network require manual updates, which can be time-consuming and error-prone in larger or dynamic environments.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1.First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image

2.Use automatic connecting cable to connect devices with others or individually use the copper straight through cable to connect the devices and use copper cable to connect the switches.

3.Configure the PCs with IPV4 Address and subnet mask according to the IP Addressing Table given below

4.Assign IP Address

PC0-192.168.30.2 (Subnet Mask 255.255.255.240)

PC1-192.168.30.3 (Subnet Mask 255.255.255.240)

PC2-192.168.10.2 (Subnet Mask 255.255.255.224)

PC3-192.168.10.3 (Subnet Mask 255.255.255.224)

5. Configuration Setting:

Router-0:

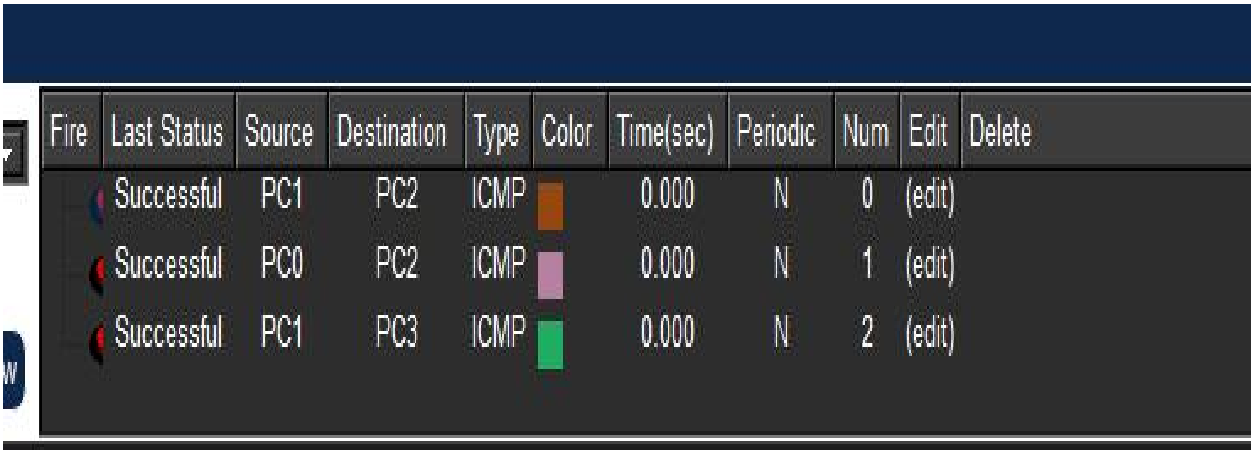
```
en
conf t
int fa0/0
ip address 192.168.30.1 255.255.255.240
no shut
exit
int fa1/0
ip address 192.168.20.1 255.255.255.252
no shut
exit
ip route 192.168.10.0 255.255.255.224 192.168.20.2
exit
```

Router-1:

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.224
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#int fa1/0
Router(config-if)#ip address 192.168.20.2 255.255.255.252
Router(config-if)#no shut
Router(config-if)#exit
Router(config)#ip route 192.168.30.0 255.255.255.240 192.168.20.1
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

6. Test Connectivity: Open a command prompt on one PC and use the "ping" command the connected to another pc.

7. Test File Sharing:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC2	ICMP		0.000	N	1	(edit)	
	Successful	PC1	PC3	ICMP		0.000	N	2	(edit)	

Conclusion: Static routing is reliable, secure, and easy to implement in small or stable networks. It offers full control over routing paths with minimal resource usage. However, it lacks flexibility and requires manual changes for any network updates. This limits its efficiency in large or frequently changing networks.

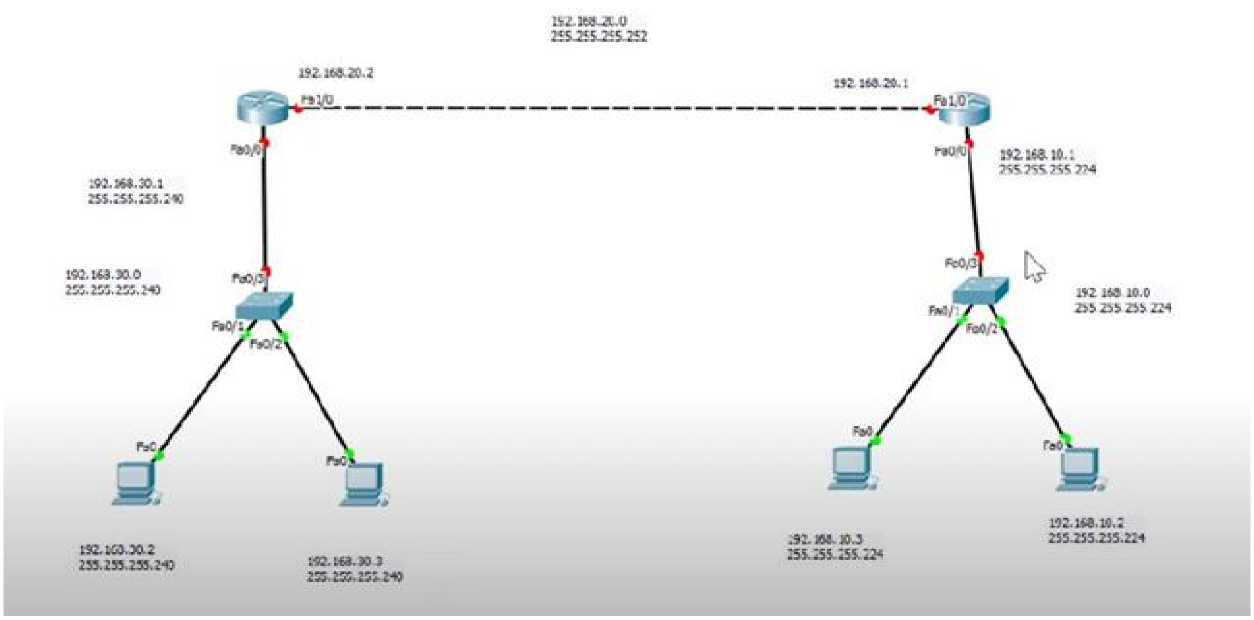
Lab Report-07

Problem: Dynamic Routing

Description: Dynamic routing is a method where routers automatically learn and update routes using routing protocols like RIP, OSPF, or EIGRP. It allows the network to adjust to changes, such as link failures or new routes, without manual intervention. This makes it highly suitable for large and complex networks. However, it consumes more bandwidth and processing power than static routing.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1.First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image

2.Use automatic connecting cable to connect devices with others or individually use the copper straight through cable to connect the devices and use copper cable to connect the switches.

3.Configure the PCs with IPV4 Address and subnet mask according to the IP Addressing Table given below

4. Assign IP Address

- PC0-192.168.30.2 (Subnet Mask 255.255.255.240)
- PC1-192.168.30.3 (Subnet Mask 255.255.255.240)
- PC2-192.168.10.2 (Subnet Mask 255.255.255.224)
- PC3-192.168.10.3 (Subnet Mask 255.255.255.224)

5. (i). Configuration Setting [**RIP Version 1**]:

Router-0:

```
en
conf t
int fa0/0
ip address 192.168.30.1 255.255.255.240
no shut
exit
int fa1/0
ip address 192.168.20.2 255.255.255.252
no shut
exit
router rip
network 192.168.30.0
network 192.168.20.0
exit
```

Router-1:

en

conf t

int fa0/0

ip address 192.168.30.1 255.255.255.240

no shut

exit

int fa1/0

ip address 192.168.20.1 255.255.255.252

no shut

exit

router rip

version 2

network 192.168.30.0

network 192.168.20.0

no auto summary

exit

(ii). Configuration Setting [**RIP Version 2**]:

Router-0:

en

conf t

int fa0/0

ip address 192.168.30.1 255.255.255.240

no shut

exit

int fa1/0

ip address 192.168.20.1 255.255.255.252

no shut

exit

router rip

version 2

network 192.168.30.0

network 192.168.20.0

no auto summary

exit

Router-1:

En

conf t

int fa0/0

ip address 192.168.20.2 255.255.255.255

no shut

exit

int fa1/0

ip address 192.168.10.1 255.255.255.224

no shut

exit

router rip

version 2

network 192.168.10.0

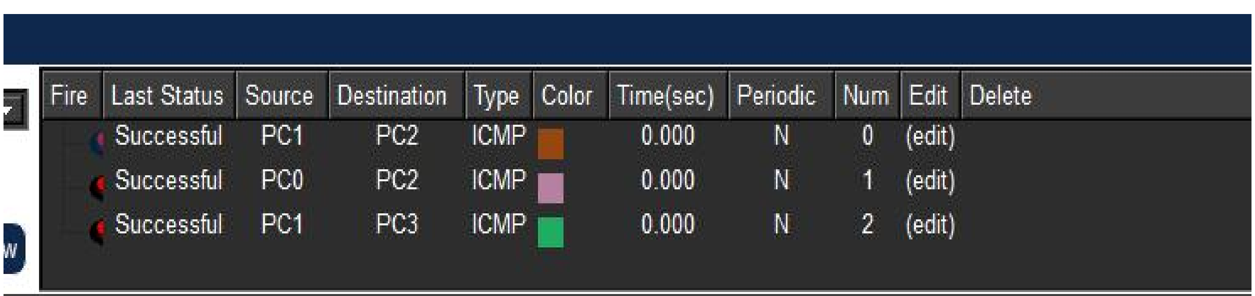
network 192.168.20.0

no auto summary

exit

6. Test Connectivity: Open a command prompt on one PC and use the "ping" command the connected to another pc.

7. Test File Sharing:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC2	ICMP		0.000	N	1	(edit)	
	Successful	PC1	PC3	ICMP		0.000	N	2	(edit)	

Conclusion: Dynamic routing automatically updates routing tables based on network changes, making it ideal for large and frequently changing networks. It reduces manual effort and adapts quickly to link failures or topology changes. Though it uses more bandwidth and resources, its flexibility and efficiency outweigh the cost. Thus, it is widely used in modern, scalable network environments.

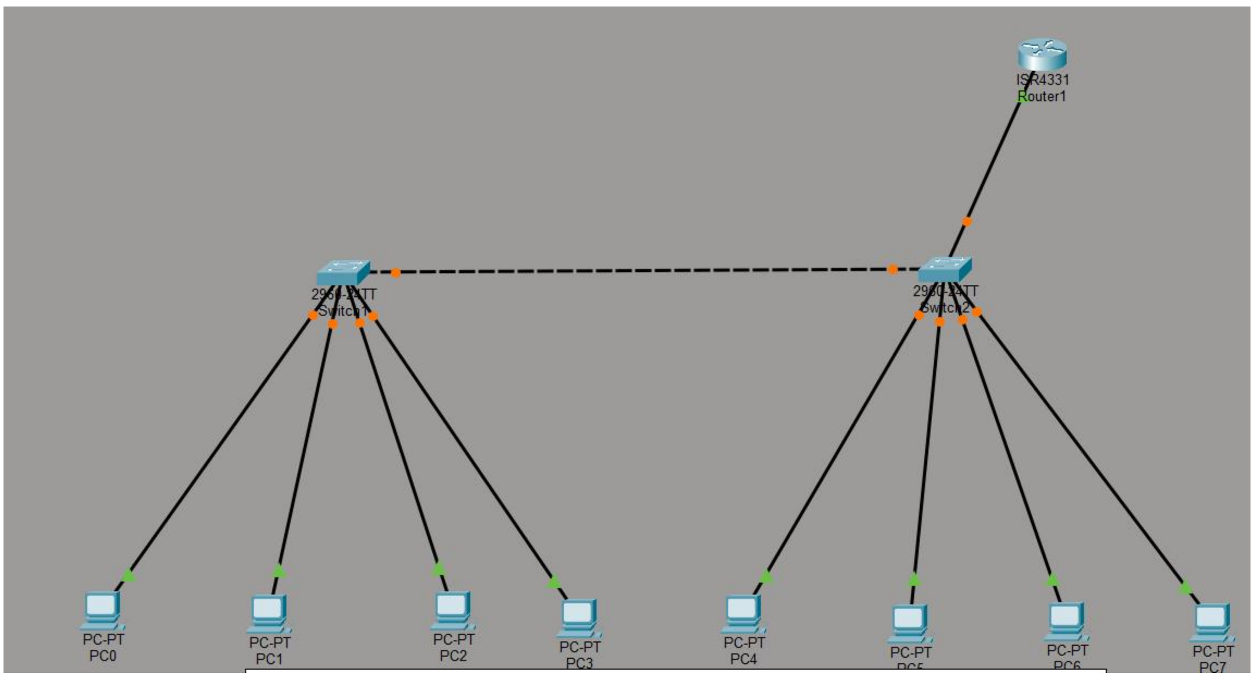
Lab Report-08

Problem: Vlan

Description: A VLAN is a logical subdivision of a physical network that groups devices based on function, department, or application, regardless of their physical location. It allows devices to communicate as if they were on the same LAN, even when connected to different switches. VLANs enhance security by isolating traffic and reducing broadcast domains, improving network performance. They are widely used in enterprise networks for better traffic management and scalability.

Necessary Tools: Cisco Packet Tracer Version 8.2.2 software

Diagram:



Working Procedure:

1.First, Open Cisco Packet Tracer in Desktop, then create a New Network Topology by End Devices (PCs), Network Devices (Switches) and Connections (Cables) as shown above the image

2.Use automatic connecting cable to connect devices with others or individually use the copper straight through cable to connect the devices and use copper cable to connect the switches.

3.Configure the PCs with IPV4 Address and subnet mask according to the IP Addressing Table given below

4.Assign IP Address

- PC0-192.168.10.1
- PC1-192.168.10.2
- PC2-192.168.20.1
- PC3-192.168.20.2
- PC4-192.168.10.3
- PC5-192.168.10.4
- PC6-192.168.20.3
- PC7-192.168.20.4

5. (i). Create a Vlan for Switch-0:

en

conf t

vlan 10

name HR

vlan 20

name IT

end

show vlan brief

Switch1

Physical Config CLI Attributes

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan breif
                     ^
% Invalid input detected at '^' marker.

Switch#
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   HR                               active
20   IT                               active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Copy Paste

(ii). Assign Port:

→ **conf t**

→ **int range fa0/1-2**

→ **switchport access vlan 10**

→ **int range fa0/3-4**

→ **switchport access vlan 20**

→ **end**

→ **show vlan brief**

→ **show interface switchport**

```
Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int range fa0/1-2
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#int range fa0/3-4
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   HR                               active    Fa0/1, Fa0/2
20   IT                               active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

(iii). Convert Dynamic to Static:

→ conf t

→ int range fa0/1-4

→ switchport mode access

→ end

→ show interface switchport

```
Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int range fa0/1-4
Switch(config-if-range)#
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (HR)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
 --More--
```

(iv). Trunking:

→ en

→ conf t

→ int range fa0/5

→ switchport mode trunk

→ end

→ show interface trunk

```
Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int range fa0/5
Switch(config-if-range)#switchport mode trunk

Switch(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/5       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/5       1-1005

Port        Vlans allowed and active in management domain
Fa0/5       1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,10,20

Switch#
```

6. (i). Create a Vlan for Switch-1:

→ en

→ conf t

→ vlan 10

→ name HR

→ vlan 20

→ name IT

→ end

→ show vlan brief

```
Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name HR
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name IT
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   HR                               active
20   IT                               active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

(ii). Assign Port:

→ conf t

→ int range fa0/1-2

→ switchport access vlan 10

→ int range fa0/3-4

→ switchport access vlan 20

→ end

→ show vlan brief

→ show interface switchport

```
Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int range fa0/1-2
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#int range fa0/3-4
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   HR                               active    Fa0/1, Fa0/2
20   IT                               active    Fa0/3, Fa0/4
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

(iii). Convert Dynamic to Static:

→ conf t

→ int range fa0/1-4

→ switchport mode access

→ end

→ show interface switchport

```
Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int range fa0/1-4
Switch(config-if-range)#
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: static access
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (HR)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
```

(iv). Trunking:

→ en

→ conf t

→ **int range fa0/5**

→ **switchport mode trunk**

→ **end**

→ **show interface trunk**

```
Switch#en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int range fa0/5
Switch(config-if-range)#switchport mode trunk
Switch(config-if-range)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/5       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/5       1-1005

Port        Vlans allowed and active in management domain
Fa0/5       1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/5       1,10,20

Switch#
```

7. **Router Configuration Setting:**

→ **en**

→ **conf t**

→ **int gig0/0**

→ **no shut**

→ **int gig0/0.1**

→ **encapsulation dot1q 10**

→ **ip add 192.168.10.5 255.255.255.0**

→ **exit**

→ **int gig0/0.2**

→ **encapsulation dot1q 20**

→ **ip add 192.168.20.5 225.225.225.0**

→**end**

```
Router> configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)# int g0/0/0
Router(config-if)#no shut
%LINK-5-CHANGED:  Interface GigabitEthernet0/0/0, changed state to up
Router(config-if)#exit
Router(config)#int g0/0/0.1
Router(config-subif)#encapsulation dot1q 10
Router(config-subif)#ip add 192.168.10.5 255.255.255.0
Router(config-subif)#exit
Router(config)#int g0/0/0.2
Router(config-subif)#encapsulation dot1q 20
Router(config-subif)#ip add 192.168.20.5 255.255.255.0
Router(config-subif)#exit
Router(config)#end
```

8. Test Connectivity: Open a command prompt on one PC and use the "ping" command the connected to another pc

9. Test File Sharing:

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	PC2	ICMP		0.000	N	0	(edit)	
	Successful	PC0	PC2	ICMP		0.000	N	1	(edit)	
	Successful	PC1	PC3	ICMP		0.000	N	2	(edit)	

Conclusion: VLAN (Virtual Local Area Network) effectively segments a physical network into multiple logical networks, improving security, performance, and manageability. In a lab environment, VLANs help simulate real-world enterprise setups by allowing multiple departments or groups to operate on separate broadcast domains without additional hardware. They reduce network congestion and provide better control over traffic. Overall, VLANs are a crucial concept in modern networking, making network design more flexible and efficient.